1A Adding and Subtracting Integers

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Evaluate each integer addition:

(a) $-14 + 9$ (b) $18 + (-25)$ (c) $-23 + (-19)$ (d) $-37 + 52$

Q2. Evaluate each integer subtraction by rewriting adjacent signs:

(a) $11 - 28$ (b) $-16 - 15$ (c) $24 - (-18)$ (d) $-33 - (-41)$

Q3. Evaluate the following multi-step directed number expressions:

(a) $-12 + 19 - (-8) - 21$ (b) $35 - (-14) + (-29) - 26$ (c) $-48 - (-15) - 17 + 34$ (d) $-105 + 68 - (-47) + (-22)$

Q4. Find the missing integer $\square$ that makes each statement true:

(a) $-15 + \square = 8$ (b) $12 - \square = 29$ (c) $\square - (-14) = -5$ (d) $-27 - \square = -11$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. At Perisher Valley in the Snowy Mountains, the temperature at $9{:}00\ \text{pm}$ was -3°C. Overnight it fell by 6°C to its minimum at $5{:}00\ \text{am}$, then rose by 14°C by $1{:}00\ \text{pm}$. Find (a) the $5{:}00\ \text{am}$ minimum temperature and (b) the $1{:}00\ \text{pm}$ temperature.

Q6. A small business overdraft account in Geelong has a balance of $-\$340$. During the week, a client EFT transfer of $\$520$ is deposited, an electricity bill of $\$265$ is debited, and a GST refund of $\$110$ is credited. What is the final account balance?

Q7. An autonomous underwater research drone in Bass Strait begins at an elevation of $-45\ \text{m}$ relative to sea level. It descends $28\ \text{m}$, rises $19\ \text{m}$ to photograph a reef, and then descends another $34\ \text{m}$. What is the drone's final elevation relative to sea level?

Q8. Lake Eyre (Kati Thanda) has a lowest point at $-15\ \text{m}$ relative to sea level, while Mount Bogong in Victoria has a summit at $1986\ \text{m}$ above sea level. Write a directed number subtraction to find the vertical height difference between the two landmarks.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Let p and q be two negative integers with $p < q$ (for example, $p = -10$ and $q = -3$). State whether $(p - q)$ is always positive or always negative, and justify your answer using a number line argument.

Q10. Each row, column, and main diagonal of a 3×3 magic square has the same magic sum $S = -6$. The top row is $(-7, 4, -3)$ and the middle row is $(2, -2, -6)$. Find the three integers in the bottom row from left to right.